

$U(1)$ Gauge Field Around a Black Hole

1 Quantum Corrections to Maxwell's Equations

1.1 Summary of the Relevant Equations

First Solution:

$$A_\mu = \{0, 0, \partial_z \Phi_1, -\partial_z \Phi_1\} \quad (1)$$

where

$$\partial_t^2 \Phi_1 - \frac{1 - \frac{2M}{r}}{1 + \frac{2\xi^2 M}{r^3}} \partial_r \left(\left(1 - \frac{2M}{r}\right) \left(1 + \frac{2\xi^2 M}{r^3}\right) \partial_r \Phi_1 \right) = -\frac{l(l+1)}{r^2} \left(1 - \frac{2M}{r}\right) \frac{1 - \frac{4\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}} \Phi_1 \quad (2)$$

The second solution:

$$A_\mu = \left\{ 0, \frac{l(l+1)\Phi_2}{r^2 \left(1 - \frac{4\xi^2 M}{r^3}\right)}, \frac{1 - \frac{2M}{r}}{1 + \frac{2\xi^2 M}{r^3}} \partial_z \partial_r \Phi_2, \frac{1 - \frac{2M}{r}}{1 + \frac{2\xi^2 M}{r^3}} \partial_z \partial_r \Phi_2 \right\} \quad (3)$$

where

$$\partial_t^2 \Phi_2 - \left(1 - \frac{2M}{r}\right) \left(1 + \frac{2\xi^2 M}{r^3}\right) \partial_r \left(\frac{1 - \frac{2M}{r}}{1 + \frac{2\xi^2 M}{r^3}} \partial_r \Phi_2 \right) = -\frac{l(l+1)}{r^2} \left(1 - \frac{2M}{r}\right) \frac{1 + \frac{2\xi^2 M}{r^3}}{1 - \frac{4\xi^2 M}{r^3}} \Phi_2 \quad (4)$$

1.2 WKB Approximation

Let Φ be either Φ_1 and Φ_2

$$\Phi(r, t) = A(r) e^{i\phi(r) - i\omega t} Y_l^m(\theta, \varphi) \quad (5)$$

The general structure of the kinds of equations we want to solve is

$$\partial_t^2 \Phi - q(r) \frac{\partial}{\partial r} \left(p(r) \frac{\partial \Phi}{\partial r} \right) = -\frac{l(l+1)}{r^2} q(r) H(r) \Phi \quad (6)$$

In the case of Φ_1

$$q(r) = \frac{1 - \frac{2M}{r}}{1 + \frac{2\xi^2 M}{r^3}}, \quad p(r) = \left(1 - \frac{2M}{r}\right) \left(1 + \frac{2\xi^2 M}{r^3}\right), \quad H(r) = 1 - \frac{4\xi^2 M}{r^3} \quad (7)$$

In the case of Φ_2

$$q(r) = (1 - \frac{2M}{r})(1 + \frac{2\xi^2 M}{r^3}), \quad p(r) = \frac{1 - \frac{2M}{r}}{1 + \frac{2\xi^2 M}{r^3}}, \quad H(r) = \frac{1}{1 - \frac{4\xi^2 M}{r^3}} \quad (8)$$

No matter which equation we are talking about, if we substitute the ansatz into the general equation what we get is

$$\begin{aligned} -\omega^2 A e^{i\phi} - q \partial_r [p(A' + iA\phi') e^{i\phi}] &= -\frac{l(l+1)}{r^2} q H A e^{i\phi} \\ \Rightarrow \quad \omega^2 A + q [p'(A' + iA\phi') + p(A'' + iA'\phi' + iA\phi'') + ip(A' + iA\phi')\phi'] &= \frac{l(l+1)}{r^2} q H A \end{aligned} \quad (9)$$

Assuming that A and ϕ are real functions

$$A'' - A\phi'^2 + \frac{p'}{p}A' = \frac{l(l+1)}{r^2} \frac{H}{p} A - \frac{\omega^2}{qp} A \quad (10)$$

$$A\phi'' + 2A'\phi' + \frac{p'}{p}A\phi' = 0 \quad (11)$$

$$\frac{1}{p} \frac{d}{dr}(pA') - A\phi'^2 = \left[\frac{l(l+1)}{r^2} \frac{H}{p} - \frac{\omega^2}{qp} \right] A, \quad \frac{d}{dr}(pA^2\phi') = 0 \quad (12)$$

If $\frac{d}{dr}(pA') \approx 0$ then

$$pA^2\phi' = C^2 \quad (13)$$

where C is a constant

$$\begin{aligned} -A \left(\frac{C^2}{pA^2} \right)^2 &= \left[\frac{l(l+1)}{r^2} \frac{H}{p} - \frac{\omega^2}{qp} \right] A \quad \Rightarrow \quad \frac{C^4}{p} = \left[\frac{\omega^2}{q} - \frac{l(l+1)}{r^2} H \right] A^4 \\ \Rightarrow \quad A &= C \left(\frac{q/p}{\omega^2 - \frac{l(l+1)}{r^2} qH} \right)^{1/4} \end{aligned} \quad (14)$$

$$\phi' = \frac{C^2}{pA^2} = \frac{1}{p} \sqrt{\frac{p}{q} \left[\omega^2 - \frac{l(l+1)}{r^2} qH \right]} = \frac{1}{\sqrt{qp}} \sqrt{\omega^2 - \frac{l(l+1)}{r^2} qH} \quad (15)$$

Putting this all together, for the $\Phi = \Phi_1$ case

$$\begin{aligned} \Phi_1 &= C_{\omega lm}^{(1)} e^{-i\omega t} Y_l^m(\theta, \varphi) \left(\frac{q/p}{\omega^2 - \frac{l(l+1)}{r^2} qH} \right)^{1/4} \exp \left[\pm i \int dr \frac{1}{\sqrt{qp}} \sqrt{\omega^2 - \frac{l(l+1)}{r^2} qH} \right] \\ &= \frac{C_{\omega lm}^{(1)} e^{-i\omega t} Y_l^m(\theta, \varphi)}{\sqrt{1 + \frac{2\xi^2 M}{r^3}}} \left(\omega^2 - \frac{l(l+1)}{r^2} \left(1 - \frac{2M}{r} \right) \frac{1 - \frac{4\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}} \right)^{-1/4} \\ &\quad \times \exp \left[\pm i \int dr \frac{1}{|1 - \frac{2M}{r}|} \sqrt{\omega^2 - \frac{l(l+1)}{r^2} \left(1 - \frac{2M}{r} \right) \frac{1 - \frac{4\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}}} \right] \end{aligned} \quad (16)$$

On the other hand, for the $\Phi = \Phi_2$ case

$$\begin{aligned} \Phi_2 = & C_{\omega lm}^{(2)} e^{-i\omega t} Y_l^m(\theta, \varphi) \sqrt{1 + \frac{2\xi^2 M}{r^3}} \left(\omega^2 - \frac{l(l+1)}{r^2} \left(1 - \frac{2M}{r}\right) \frac{1 + \frac{2\xi^2 M}{r^3}}{1 - \frac{4\xi^2 M}{r^3}} \right)^{-1/4} \\ & \times \exp \left[\pm i \int dr \frac{1}{|1 - \frac{2M}{r}|} \sqrt{\omega^2 - \frac{l(l+1)}{r^2} \left(1 - \frac{2M}{r}\right) \frac{1 + \frac{2\xi^2 M}{r^3}}{1 - \frac{4\xi^2 M}{r^3}}} \right] \end{aligned} \quad (17)$$

2 Transmission Amplitude

$$U(r) = \frac{l(l+1)}{r^2} \left(1 - \frac{2M}{r}\right) \quad (18)$$

$$U_1(r) = \frac{l(l+1)}{r^2} \left(1 - \frac{2M}{r}\right) \frac{1 - \frac{4\xi^2 M}{r^3}}{1 + \frac{2\xi^2 M}{r^3}}, \quad U_2(r) = \frac{l(l+1)}{r^2} \left(1 - \frac{2M}{r}\right) \frac{1 + \frac{2\xi^2 M}{r^3}}{1 - \frac{4\xi^2 M}{r^3}} \quad (19)$$

The transmission amplitude

$$T_l(\omega) = \quad (20)$$

where

$$\theta_l(\omega) = \int_{r_1}^{r_2} \frac{dr}{1 - \frac{2M}{r}} \sqrt{U(r) - \omega^2} \quad (21)$$

Find the maximum of $U(r)$

$$\left. \frac{dU(r)}{dr} \right|_{r=r_{max}} = -\frac{2l(l+1)}{r_{max}^3} \left(1 - \frac{3M}{r_{max}}\right) = 0 \quad \Rightarrow \quad r_{max} = 3M \quad (22)$$

$U(r)$ at maximum

$$U_{max} = U(r_{max}) = \frac{l(l+1)}{9M^2} \left(1 - \frac{2}{3}\right) = \frac{l(l+1)}{27M^2} \quad (23)$$

As l increases the top of the barrier increases as well. For a given frequency ω there will be some l_c at which $\omega^2 < U_{max, l_c}$. The l_c at which this happens is

$$\omega^2 = U_{max, l_c} = \frac{l_c(l_c+1)}{27M^2} \quad \Rightarrow \quad l_c(l_c+1) = 27\omega^2 M^2 \quad (24)$$

The expression for the absorption cross section is

$$\sigma_{abs} = \frac{\pi}{2\omega^2} \sum_{l=1}^{\infty} (2l+1) \left[2 - R_l^{(+)} - R_l^{(\times)} \right] = \frac{\pi}{2\omega^2} \sum_{l=1}^{\infty} (2l+1) \left[T_l^{(+)} + T_l^{(\times)} \right] \quad (25)$$

where $R_l^{(+)}$ and $R_l^{(\times)}$ are the reflection probabilities for the two polarization states, and $T_l^{(+)}$ and $T_l^{(\times)}$ are the corresponding transmission probabilities. In this case $T_l^{(+)} = T_l^{(\times)} = T_l$. Note that for $\omega^2 < U_{max}$ the transmission probability is essentially equal to zero. On the other hand, if $\omega^2 > U_{max}$ the transmission probability is virtually 100%. So

$$T_l = \begin{cases} 1 & \text{if } l < l_c \\ 0 & \text{if } l > l_c \end{cases} \quad (26)$$

Thus,

$$\sigma_{abs} = \frac{\pi}{\omega^2} \sum_{l=1}^{\infty} (2l+1) T_l = \frac{\pi}{\omega^2} \sum_{l=1}^{l_c} (2l+1) = \frac{\pi l_c (l_c + 2)}{\omega^2} \quad (27)$$

Taking (24)

$$l_c(l_c + 1) = 27\omega^2 M^2 \quad \Rightarrow \quad l_c = \frac{\sqrt{1 + 108\omega^2 M^2} - 1}{2} \quad (28)$$

Substituting

$$\sigma_{abs} = \frac{\pi}{2\omega^2} (\sqrt{1 + 108\omega^2 M^2} - 1 + 54\omega^2 M^2) \approx 54\pi M^2 - 729\pi\omega^2 M^4 \quad (29)$$

$$\begin{aligned} & \left. \frac{dU_1(r)}{dr} \right|_{r=r_{max}+\delta r} \\ = & - \frac{2l(l+1)}{r^4} \frac{(r^6 - 8\xi^4 M^2)(r - 3M) + (24M - 11r)r^3 \xi^2 M}{(r^3 + 2\xi^2 M)^2} \Big|_{r=3M+\delta r} = 0 \\ \Rightarrow & (r^6 - 8\xi^4 M^2)(r - 3M) + (24M - 11r)r^3 \xi^2 M \Big|_{r=3M+\delta r} = 0 \\ \Rightarrow & -243\xi^2 M^5 + (729M^6 - 540\xi^2 M^4 - 8\xi^4 M^2)\delta r \approx 0 \\ \Rightarrow & \delta r \approx \frac{243\xi^2 M^3}{729M^4 - 540\xi^2 M^2 - 8\xi^4} \approx \frac{\xi^2}{3M} \end{aligned} \quad (30)$$

$$U_1(3M + \delta r) \approx \frac{l(l+1)}{27M^2} - \frac{2l(l+1)\xi^2}{243M^4} = \frac{l(l+1)}{27M^2} \left(1 - \frac{2\xi^2}{9M^2} \right) \quad (31)$$

$$\omega^2 = \frac{l_c(l_c + 1)}{27M^2} \left(1 - \frac{2\xi^2}{9M^2} \right) \quad \Rightarrow \quad l_c(l_c + 1) = \frac{27\omega^2 M^2}{1 - \frac{2\xi^2}{9M^2}} \quad (32)$$

$$\sigma_{abs}^{(+)} = \frac{\pi l_c^{(+)} (l_c^{(+)} + 2)}{2\omega^2} \approx 27\pi M^2 + 6\pi\xi^2 = \frac{27}{4}\pi r_s^2 + 6\pi\xi^2 \quad (33)$$

Similarly

$$\sigma_{abs}^{(\times)} = \frac{\pi l_c^{(\times)} (l_c^{(\times)} + 2)}{2\omega^2} \approx 27\pi M^2 - 6\pi\xi^2 = \frac{27}{4}\pi r_s^2 - 6\pi\xi^2 = \frac{27\pi}{4} \left(r_s^2 - \frac{8}{9}\xi^2 \right) \quad (34)$$